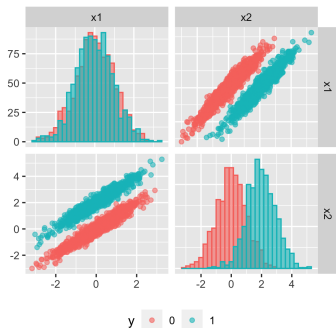
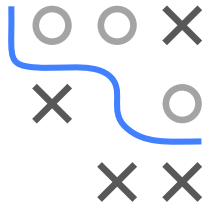


Introduction to Machine Learning

Feature Selection

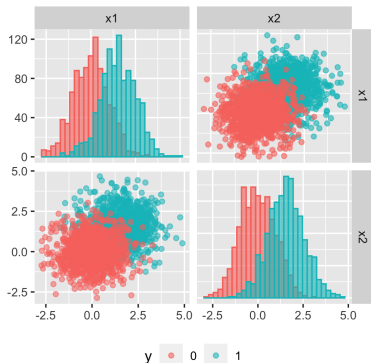
Feature Selection: Filter Methods (Examples and Caveats)



Learning goals

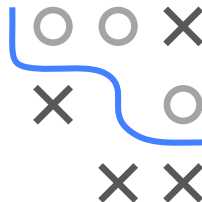
- Understand how filter methods can be misleading
- Understand how filters can be applied and tuned

FILTER METHODS CAN BE MISLEADING



ρ_{ACC} of log. reg. classifier with:

- feature x_1 : 0.76
- feature x_2 : 0.78
- both features: 0.85



Information gain from presumably redundant variables. 2 class problem with indep features. Each class has Gaussian distribution with no covariance. While filter methods suggest redundancy, combination of both vars yields improvement, showing indep vars are not truly redundant. For further details, see [► Guyon and Elisseeff 2003](#).

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- Performance of classif.kknn on dataset har
- classif.mbrrier
- Number of Features
- 210 0.041
- 288 0.042
- F-test ● information gain

Feature Score and log₁₀ reg model performance. Dataset: SPAM

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